



Cambridge International AS & A Level

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FURTHER MATHEMATICS

9231/12

Paper 1 Further Pure Mathematics 1

May/June 2026
2 hours

You must answer on the question paper.

You will need: List of formulae (MF19)

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen. Do **not** use correction fluid or tape.
- Do **not** write on any bar codes.
- If additional space is needed, you should use the lined page at the end of this booklet; the question number or numbers must be clearly shown.
- You should use a calculator where appropriate.
- You must show all necessary working clearly; no marks will be given for unsupported answers from a calculator.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.

INFORMATION

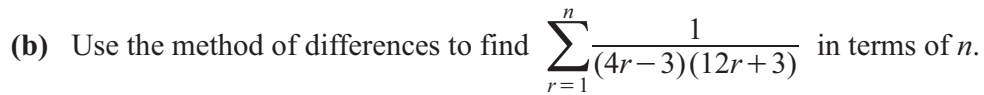
- The total mark for this paper is 75.
- The number of marks for each question or part question is shown in brackets [].

 This document has **20** pages. Any blank pages are indicated.

- $$\sum_{r=1}^n (4r-3)(12r+3) = an^3 + bn^2 + cn,$$

[3]

[illegible]



[4]

[illegible]

(c) State the value of $\sum_{r=1}^{\infty} \frac{1}{(4r-3)(12r+3)}$.

[1]

.....

[illegible]

* 0000800000005 *



5

Turn over



- (a) Show that $\mathbf{M} = \begin{pmatrix} k & 2 \\ 0 & 1 \end{pmatrix}$.

[illegible]



- (b) The transformation represented by $\mathbf{M}$ has a line of invariant points.

Find, in terms of k , the equation of this line.

[3]

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The square S in the x - y plane is transformed by $\mathbf{M}$ onto the parallelogram P .

- (c) Given that the area of S is half of the area of P , find the possible values of k .

[2]

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4 The cubic equation $5x^3 - 2x + 1 = 0$ has roots α, β, γ .

(a) Find a cubic equation whose roots are $\frac{1}{\alpha^2}, \frac{1}{\beta^2}, \frac{1}{\gamma^2}$. [3]

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(b) Find the value of $\frac{1}{\alpha^4} + \frac{1}{\beta^4} + \frac{1}{\gamma^4}$. [3]

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[4]

[illegible]

(a) Find the shortest distance between l_1 and l_2 . [5]

[illegible]



- Find the acute angle between Π_1 and Π_2 . [6]

[illegible]



6 The curve C has equation $y = \frac{x^2 - x + 2}{4x + 4}$.

(a) Find the equations of the asymptotes of C .

[3]

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(b) Find the coordinates of any stationary points on C .

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(c) Sketch C .

[3]

(d) Sketch the curve with equation $y = \left| \frac{x^2 - x + 2}{4x + 4} \right|$.

[1]

[4]

[illegible]



(a) Show that, at the point on C furthest from the initial line,

$$\cos \theta - \frac{\sin \theta}{\cos 2\theta (\sin 2\theta + \cos 2\theta)} = 0$$

and verify that this equation has a root between 0.5 and 0.6.

[6]

[illegible]

[2]

$$= \tan 2\theta \text{ to show that the area enclosed by the curve is } \frac{1}{4}a \int_0^\infty \frac{1}{(1+v)(1+v^2)} dv$$

[9]

[illegible]



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Handwriting practice area with 20 horizontal dotted lines.



[illegible]



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